

"balance" or "zero set" condition on the meter. Then the resistors are exposed to the reference sample and gas sample.

If the gas sample contains elements having a different thermal conductivity than the reference sample, the bridge will be unbalanced. In some cases, the meter reads out in terms of the thermal conductivity, whereas in other analyzers, the indication is on a *go-no go* or *good-bad* basis.

Measurement of Thermal Conductivity Using Thermistors. For this application two thermistors are connected in the adjacent arms of a Wheatstone bridge as shown in Fig. 29'99. The bridge supply is high enough to raise the thermistors above ambient temperature, typically to about 150°C . One thermistor is mounted in a static area to provide temperature compensation while the other is placed in the medium to be measured.

NUCLEAR INSTRUMENTATION

29'49 Introduction. A lot of instrumentation is needed to measure the amount of radioactivity encountered in nuclear research work and for tracking of radiated alpha, beta, and gamma particles. The energy of these particles is absorbed by other molecules, producing chemical changes, ionization, or other reactions that produce visual evidence of the presence of other particles. These processes are used to detect, count and trace the movement of such particles.

29'50. Types of Radiations. Radioactive materials emit nuclear radiations which mainly consist of four parts:

1. **Alpha Particles.** The alpha particle is a helium nucleus consisting of two protons and two neutrons. It has a mass of 6.2×10^{-27} kg and carries a positive charge of 3.2×10^{-9} C. The alpha particle is the heaviest and the slowest, travelling at speeds of from 1/10 to 1/100 of the velocity of light. Since it has the greatest mass and slowest speed, it is least penetrating. It penetrates about 0.02 mm of aluminium foil.

2. **Beta Particles.** The beta particle is an electron with a mass of 9.03×10^{-31} kg and a charge of 1.6×10^{-19} C. Since beta particle is much lighter than alpha particle and has a speed approximately equal to that of light, it can penetrate much farther than alpha particle—about 100 times the penetration of alpha particle.

3. **Gamma Particles** Gamma particles or gamma rays, are an electromagnetic radiation with a wavelength of approximately 0.03 to 3 angstrom. Gamma particles travel at the speed of light, and because of their shorter wavelengths, have very great penetrating power. Gamma rays can penetrate about 20 cm through a mass of lead.

4. **Neutrons.** The mass of the neutron is intermediate between that of alpha and beta particles, having the same mass as a proton. The speeds of neutrons may be high or low depending upon the source from which they are emitted. Because neutron has no charge it is very difficult to detect. However, its penetrating power is small.

The above mentioned properties and other properties of these particles must be considered in the methods and processes for their detection and measurement. In the following pages, a few methods of detection and measurement of the above mentioned particles are described.

29'51. Geiger Muller Tube. The Geiger Muller tube or simply Geiger tube are frequently used for detection and measurement of alpha, beta and gamma rays. This tube is a diode, consisting of a cathode which is a long metal cylinder and an anode which is a fine wire running through the centre of the cylinder. Both of them are mounted in a thin walled, air tight, glass envelope, sealed by an extremely thin window through which radiations may pass at one end (See Fig. 29'100). The air is evacuated from the envelope and a small amount of an inert gas, such as Argon, is added.

The voltage of the battery is kept

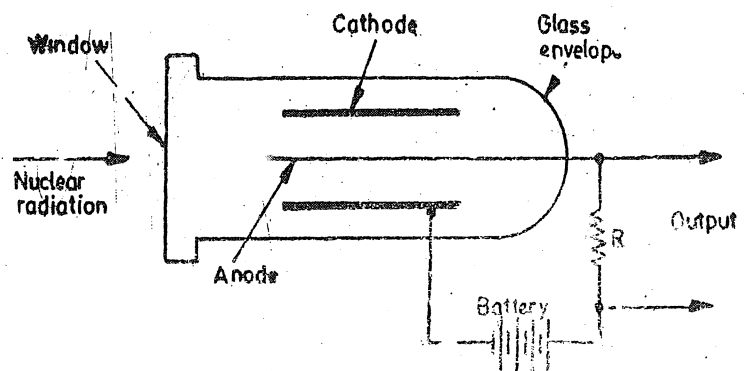


Fig. 29'100. Geiger Muller tube.

MEASUREMENT OF NON-ELECTRICAL QUANTITIES

below the ionization potential of the gas. Now if radiation penetrates through the window and enters the envelope, it would ionize some of the gas atoms. The resulting negative ions go towards the anode and positive ions towards the cathode. In their passage, the ions collide with some of the gas atoms, causing them to be ionized in turn. This process continues till whole of the gas atoms are ionised. In fact complete ionization takes place in no time. A pulse of current thus flows through the tube. This current flows through the resistance, R , which is connected in the anode circuit. The resulting voltage drop across R is the output voltage.

Once ionization is established, it is expected to continue indefinitely. However, since resistor R , which has a high resistance, is connected in series with the anode and the battery, when the gas ionizes a heavy current flows through resistance causing a large voltage drop. This voltage drop is sufficient to reduce the anode voltage below the ionization potential and therefore ionization ceases.

As long as the gas is not ionized no current flows through the resistance R and hence there is no output. Thus when the gas deionizes, the current ceases to flow through resistance R and so there is no output.

Now if radiation again enters the tube, the ionization starts and a current again flows through resistance R giving an output voltage. This way a series of alpha or beta particles or bursts of gamma rays, cause a series of current pulses to pass through the anode circuit of the tube and hence through resistance R . The output pulses from resistance R may be amplified and registered by a counting device. By counting the number of pulses, we can know the number of particles entering the Geiger Müller tube in a particular interval of time. The number counted gives the intensity of radiation. The pulses may be stored in some cases and the total count may be calibrated directly in terms of radio-activity.

The counting rate of this tube is seldom greater than 10^3 counts per second.

29'52. Ionization Chamber. This is quite similar to the Geiger Müller tube as shown in Fig. 29'101. The ionization chamber is a metal cylinder (the outer electrode) sealed at one end by a window. The other end of the cylinder is closed, and a metal rod (the centre electrode) protrudes outside the cylinder on this end.

The central electrode is grounded through resistance R . Resistor R has a very high resistance. A large positive potential is applied to the outer electrode. The chamber contains a gas which may be air, carbon dioxide, nitrogen, argon or methane.

The action of this ionization chamber is similar to that of Geiger Müller tube.

Let us consider that the chamber is filled with air. When a nuclear radiation enters the chamber through the window, some of the atoms of the air are ionized, the resulting ions travel

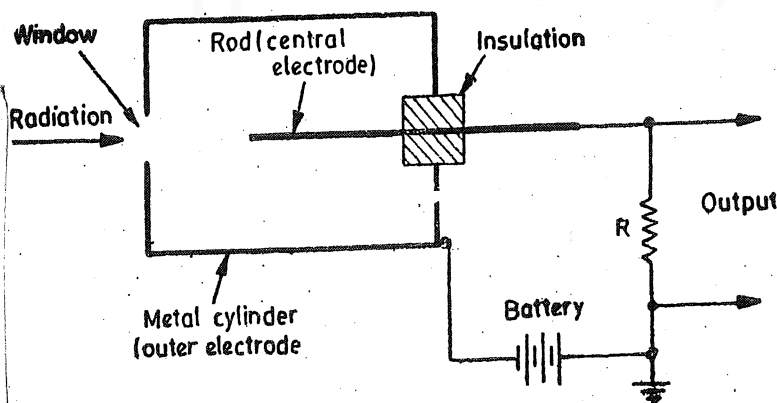


Fig. 29'101. Ionization chamber

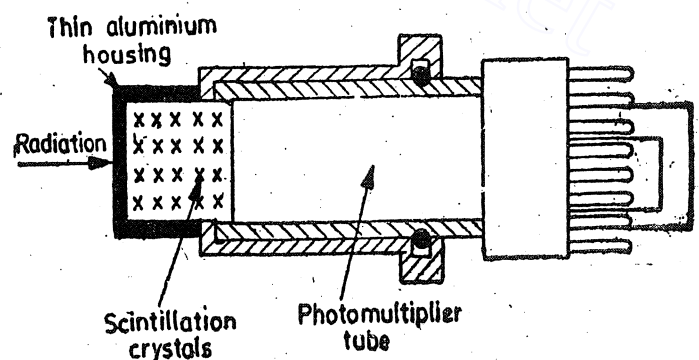


Fig. 29'102. Scintillation counter.

towards their respective electrodes i.e. positive ions towards the central electrode and the negative ions towards the outer electrode. In the process of rushing towards the electrodes, these ions collide with other atoms of air causing ionization of these atoms. The action is instantaneous.

As the positive ions collect on the central electrode (which is at ground potential), the d.c. potential of this electrode is increased. This increased potential, which is the output signal of the chamber, may then be amplified and measured. Since the increase in this potential is a function of the number of radiation particles that have entered the chamber, the intensity of the nuclear radiation may be determined. The main use of this chamber, is to detect and measure alpha particles.

29.53. Scintillation Counters. Certain crystals such as zinc sulphide, sodium iodide, anthracene and naphthalene etc., produce a brief flash of light each time they are struck by an alpha or beta or gamma particles. The production of a flash of light by striking the crystals mentioned above, with alpha or beta or gamma rays is called **Scintillation**.

This principle is employed in a nuclear radiation sensor called "**Scintillation Counter**".

The construction of a scintillation counter is shown in Fig. 29.102. Now every time a radiation strikes the scintillation crystals, a tiny flash of light is produced. The flash of light is amplified by a photomultiplier incorporated in the scintillation counter. Thus each particle produces a pulse of anode current at the output of the photomultiplier. The output of the photomultiplier is applied to a high speed electronic counter which counts each tiny flash generated by the scintillation crystals. Thus by counting the number of pulses in a given period of time, the intensity of radiation may be detected.

The advantages of a scintillation counter are:

(i) This is more sensitive than the Gieger Muller tube and hence can detect lower levels of radiation.

(ii) The crystals used in this device produce a flash of light when struck by X-rays. Hence this transducer may be used to detect X-rays as well.

(iii) Counting rate is very fast. Pulses of as short a duration as 2 ns may easily be recorded.

29.54. Solid State Ionization Sensing Elements. A system for counting the number of particles or photons whose energy level exceeds about 1,000 electron volt is built by using extrinsic semi-conductors mounted so that the radiation ionizes an atom in the crystal, and a pulse appears in the output circuitry. Operation is normally at liquid nitrogen temperatures. Counting rates as high as 10^6 per second are possible.

29.55. Detection of Neutrons. The neutrons do not carry any charge and hence are very difficult to detect. Thus we have to adopt indirect methods to detect them. Neutrons detecting methods require an arrangement where alpha or beta rays particles are produced by neutrons and these particles detected by the methods described above.

An indirect method of detection of neutrons is described below:

A neutron is made to strike the atom of a certain isotope of uranium. The uranium atom splits into two or more parts on account of fission. This releases, in the process, nuclear radiation in the form of alpha, beta and gamma rays, which may be detected by methods described above. Thus, by measuring the intensity of the radiations, the intensity of neutron beam producing them may be detected.

Only slow moving neutrons will produce fission of uranium atoms. Where fast moving neutrons are to be detected, a layer of paraffin is put on top of the uranium layer of detector. The paraffin converts the fast moving neutrons to the slow moving neutrons required for fission.

UNSOLVED PROBLEMS

1. A strain gauge having a resistance of $350\ \Omega$ and a gauge factor of 2 is connected in series with a ballast resistance across a 10 V supply. The ballast resistance is designed to give maximum sensitivity. The gauge is subjected to a dynamic strain of $(10 + 20 \sin 314 t)$ microstrain.